# Statistics

Definition –

A branch of applied maths that involves the collection description, presentation, analysis and Intereprtation of numerical data. is called statistics.

## ✱ Type of Data –

① structured data

② Unstructured data

## // Stages of stats –

① collection of data

② Organizing of data

③ Presentation of data

④ Analysis of data

⑤ Intereprefation of data

☆ Type of stats -

① Descriptive stats

Available data sample or population on it we perform action like analysing. discribe, summerize. it called descriptive stats.

## ② Inferential stats -

On describe data we perform interepetation like hypothesis testing on the data for example Z-test, T-test, F-test, chi-squere test. is called inferential statistics

① Descriptive stats.

① Univariate Des. stats.

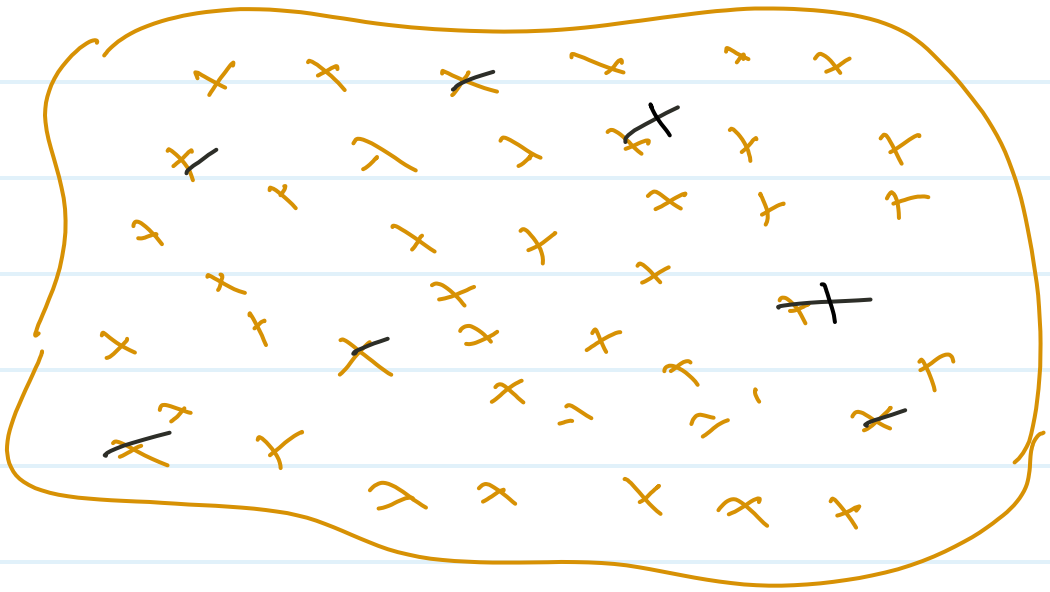
② Bivariate Des. stats.

③ Multi variate Des. stats.

# ✱ Population & Sample

Entire data available is call population.

Population = N



Sample = n

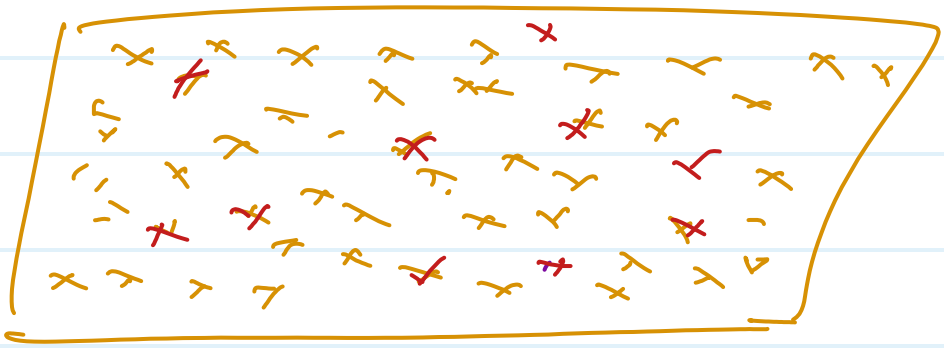
## ✱ Sampling methods –

// Type of Sampling method

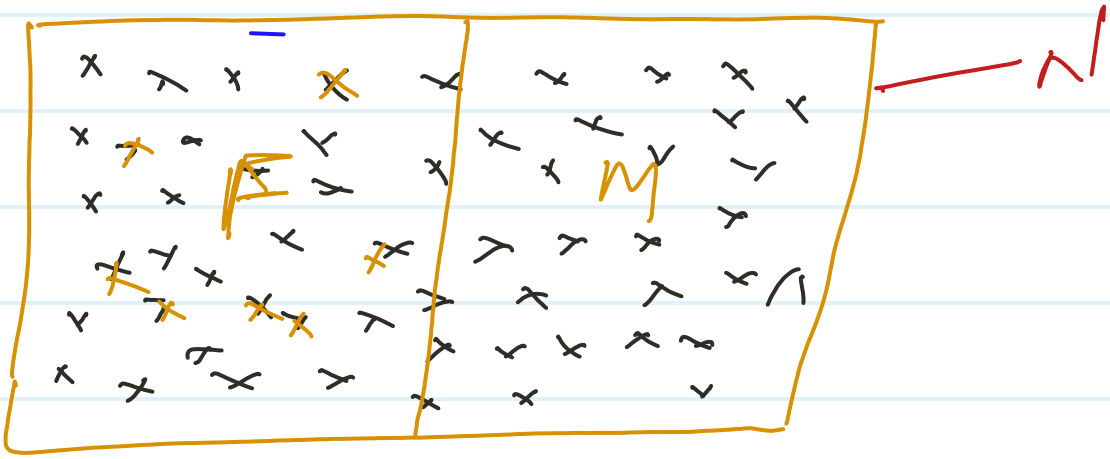
1. Probability Sampling
2. Non-probability sampling.

# Probability sampling

## ① Simple random Sampling.–



## ② stratified Sampling




> 18

Gender.

## ③ Systematic Sampling.-

$n^{th}$ select

population - 500
sample - 50

$$= \frac{50\cancel{0}}{5\cancel{0}}$$

$$= 10$$

$> 18$

$10^{th}$, $20^{th}$, $30^{th}$

[ ✓, , , ✓, , , ✓, , , ✓, , , ✓, , , ✓, , ]

## ② Non-probability Sampling -

### ① Convinience Sampling -

Research -

# How to calculate sample size?

## * Cochran Formula

$$n_0 = \frac{Z^2 pq}{e^2}$$ – infinite population

$$n = \frac{n_0}{1 + \frac{(n_0 - 1)}{N}}$$ – Required population.

$e$ = margin of error 5%

$p$ = Population proportion 50%

$q$ = $1 - p$

$Z$ = value from Z-table

| C.I. | | $\alpha$ |
|---|---|---|
| 90 % | - | 0.1 |
| 95 % | - | 0.05 |
| 99 % | - | 0.01 |

0.975

<u>Z-table</u>

$$\alpha = 0.05$$

$$1 - 0.05 = 0.95$$

$$Z = 1.6 + 0.05$$

$$\boxed{Z = 1.65}$$

$$= \frac{(1.65)^2 \times 0.5 \times (1 - 0.5)}{(0.05)^2}$$

$$= \frac{2.72 \times 0.5 \times 0.5}{0.025}$$

$$n_0 = \frac{0.68}{0.025} = \underline{\underline{27.2}}$$

Required population $N = 10000$

$$n = \frac{n_0}{1 + \frac{(n_0 - 1)}{N}}$$

$$= \frac{27.2}{1 + \frac{(27.2 - 1)}{10000}}$$

$$\Rightarrow \underline{\underline{27.12}}$$

28 / 27

If population = 10000

Sample = 28

# Descriptive stats -

1. measure of Center tendency
2. measure of Dispersion or variation
3. measure of position
4. measure of shape

## ☆ Measure of Center Tendency

1. mean -

[2, 3, 4, 5, 6]

$$\text{mean} = \frac{2+3+4+5+6}{5} = \frac{20}{5}$$

$$= 4$$

population mean $= \mu$

Sample mean $= \bar{x}$

② median –

I-case [6, 2, 4, 5, 3]

data sort

[2, 3, 4, 5, 6]

↑

median = 4

II-Case

[2, 3, 5, 7, 8, 9, 11, 13]

$$\text{median} = \frac{7+8}{2} \Rightarrow 7.5$$

③ mode –

(i) uni-modal
(ii) Bi-modal
(iii) multi-modal

① [2, 3, 4, 5, 5, 6]

mode = 5

② [2, 2, 4, 5, 6, 6]

mode = 2, 6

③ [2, 2, 4, 4, 5, 5, 6, 7, 8]

mode = 2, 4, 5

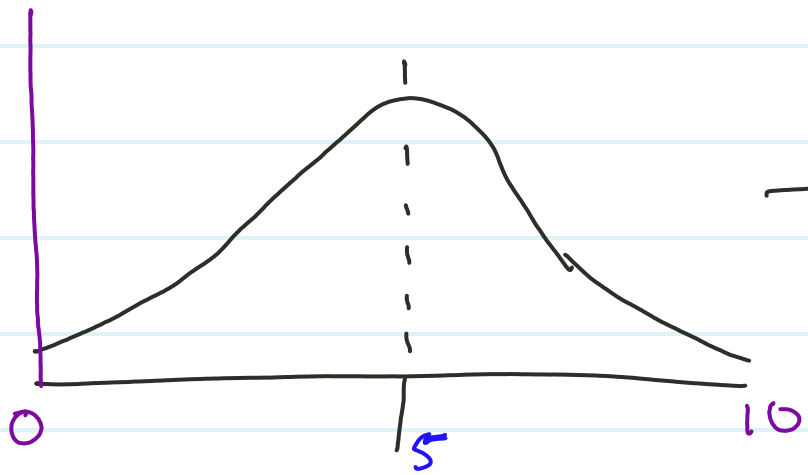
→ main application

→ outlier handeling

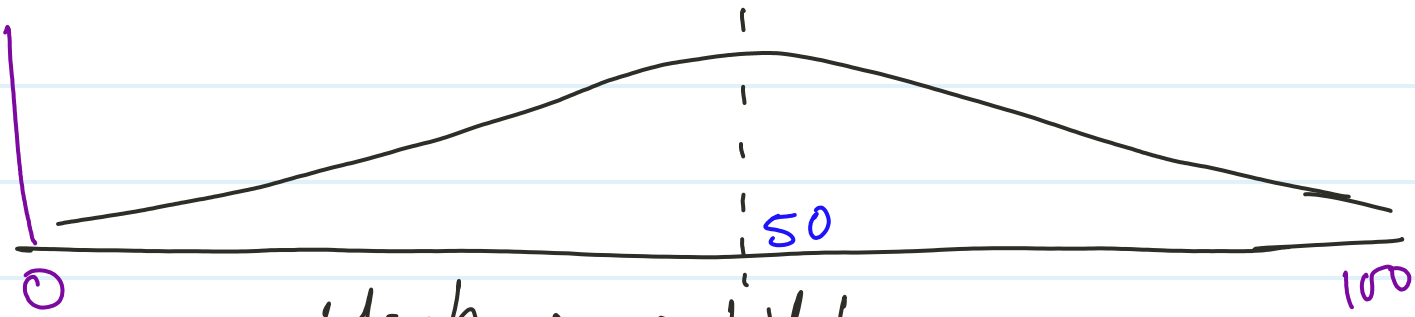
→ missing value handel

| | age | salary | Result |
|---|---|---|---|
| | 20 | 10 | P |
| | 18 | 15 | F |
| Null | — | — | P |
| | 15 | 30 | F |
| | 22 | — | — |
| Null | — | 35 | — |
| | 27 | 120 | F |

# * measure of Dispersion or variance




— low variability




High variability

① mean absolute deviation -

The mean absolute devi. of a dataset is the avg. distance b/w each data point and the mean.

$$\Rightarrow \quad \frac{1}{N} \sum_{i=1}^{n} |x_i - \bar{x}|$$

Data - [10, 15, 15, 17, 18, 21]

$$\text{mean} = \frac{96}{6} = 16$$

$$|10-16| = |-6| = 6$$
$$|15-16| = |-1| = 1$$
$$|15-16| = 1$$
$$|17-16| = 1$$
$$|18-16| = 2$$
$$|21-16| = 5$$
$$16$$

$$\Rightarrow \frac{16}{6} \Rightarrow 2.67$$

② Variance -

It tells the degree of spread in dataset.

High variability - datapoint spread widely

low variability - data point close to mean.

population $\sigma^2 = \frac{1}{N} \sum_{i=1}^{N} (x_i - \mu)^2$

sample $s^2 = \frac{1}{n-1} \sum_{i=1}^{n} (x_i - \bar{x})^2$

*Note - $n-1$ is a degree of freedom

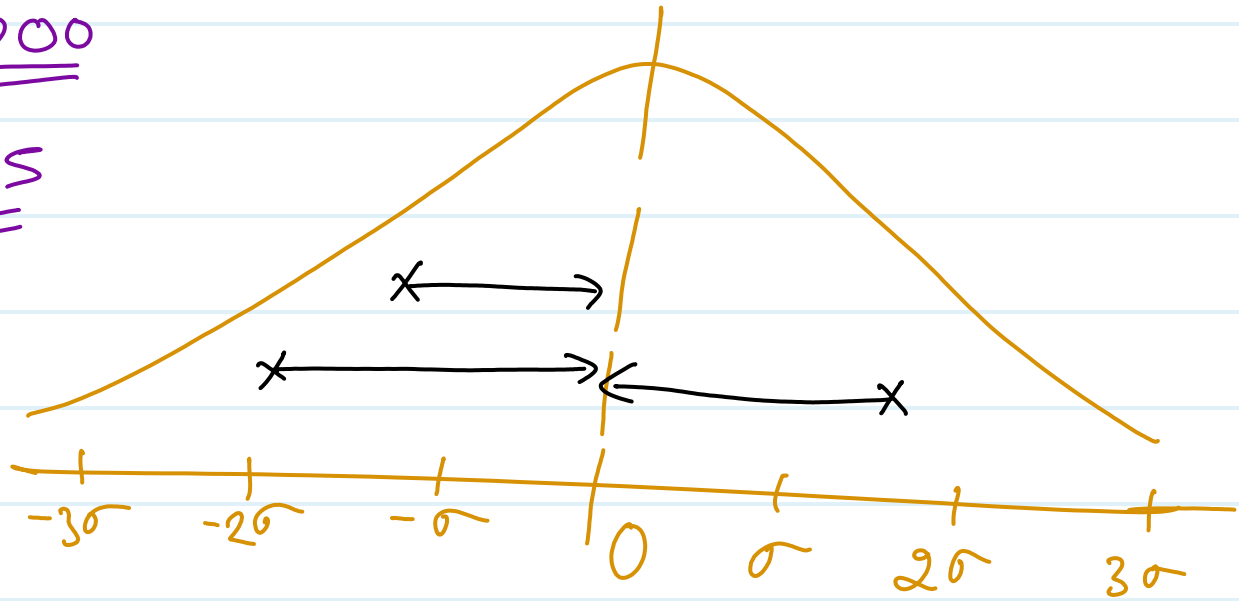
OR

Besils correction

For keep away result from biased.

200

95



## ③ standard deviation -

The square root of variance is called st. dev..

The farthes the data points from the higher the deviation

Pop $\sigma = \sqrt{\frac{1}{N} \sum_{i=1} (x_i - \mu)^2}$

Sample $S = \sqrt{\frac{1}{n-1} \sum_{i=1} (x_i - \bar{x})^2}$

## ③ Range -

[1, 2, 5, 6, 11, 15, 19, 25, 30]

max - 30

min - 1

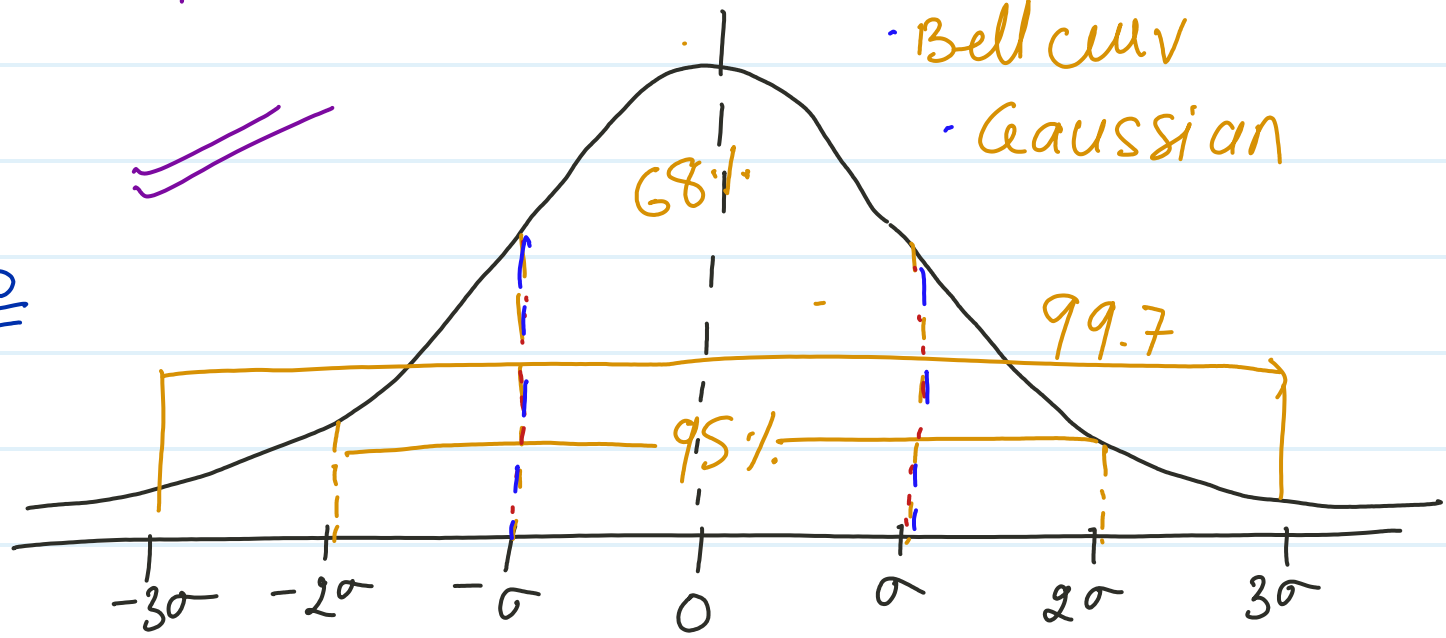
range = max - min

= 30 - 1 = 29

# * Emperical Rule

- Normal
- Bell curv
- Gaussian

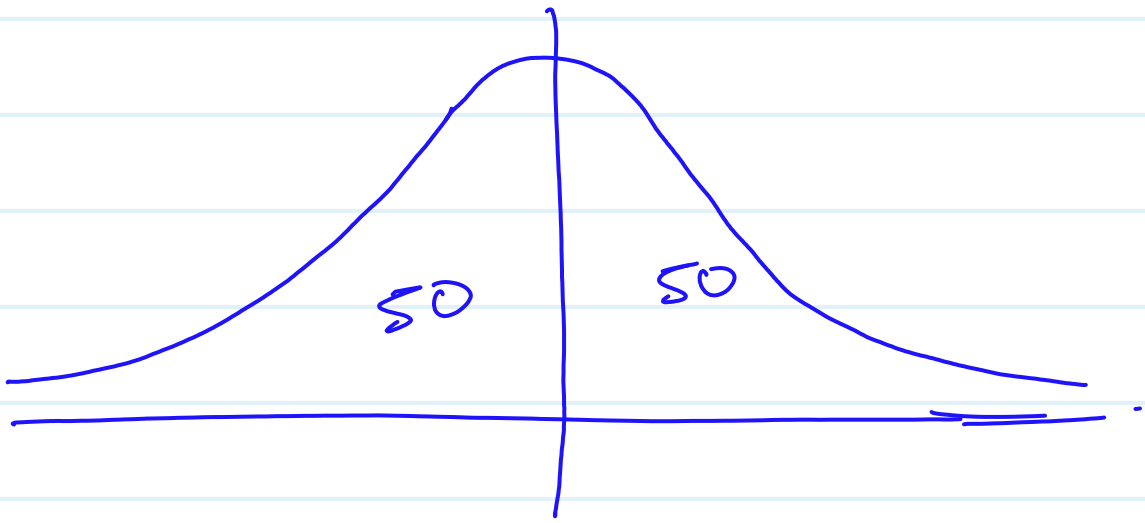
100




Normal Distribution Curv

68 — 95 — 99.7

σ 2σ 3σ



$$\Rightarrow \frac{1}{8} \times 0.7$$

$$\left[-3.6 - 1.6 + 1.4 - 4.6 + 3.4 + 6.4 + 11.4 - 6.6 - 5.6\right]$$

$$\Rightarrow \frac{0.6}{8} \qquad \frac{9.8}{8}$$

$$\underline{0.75} \qquad \underline{1.22}$$

$$\Rightarrow \sqrt{\frac{1}{8} \times \frac{0.6}{8}}$$

$$\Rightarrow \sqrt{0.75} \Rightarrow \underline{\underline{0.86}}$$